

Chokh-e-Dekha — System Requirements (v1.0)

Prepared for: People's Republic of Bangladesh

Project Type: Civic Tech Platform (Web + Mobile)

Primary Themes: Civic Tech & Public Engagement; Local Governance & Participation; Anti-Corruption Tech & Data; Legal & Justice Access

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1) Vision & Problem Statement

Vision: A national, citizen-powered platform that turns everyday civic problems into tracked cases and measurable improvements — reducing friction between people and local/state authorities, deterring corruption, and improving service delivery through transparent workflows, open data, and AI-assisted triage.

Problems today:

- Fragmented grievance systems; citizens don't know where to report and what happens next.
- Low trust due to opaque status, long resolution time, and limited feedback loops.
- Corruption and petty rent-seeking often go unreported for fear of reprisal.
- Local government lacks actionable dashboards and prioritization tools.
- Legal awareness is low; citizens don't know their rights, remedies, or supporting documentation required.

Solution: Chokh-e-Dekha (CeD) provides:

1. **Report** → **Case** → **Resolution** pipeline with SLA tracking and public status boards.

2. **Anti-corruption reporting** with safe, pseudonymous modes and evidence chain-of-custody.
3. **Legal/justice assistance:** guided flows, rights explainers, referral to legal aid, RTI templates.
4. **Local governance participation:** participatory budgeting, micro-surveys, neighborhood assemblies, and open datasets.
5. **AI-assisted triage:** dedupe reports, detect urgency, route by jurisdiction, summarize evidence, flag risk.

2) Alignment to Themes

- **Civic Tech & Public Engagement:** Citizen reporting; town-hall polls; community watch; neighborhood feeds; participatory budgeting; public status dashboards; feedback after resolution.
- **Local Governance & Participation:** Ward/union-level case queues, service SLAs, noticeboards, council agenda builder, geospatial tasking for city engineers; offline kiosks.
- **Anti-Corruption Tech & Data:** Secure whistleblower box, heatmaps of informal payments, anomaly detection in contractor payments & repetitive fault patterns; open corruption indicators.
- **Legal & Justice Access:** Rights explainer, guided complaint drafting, RTI & GD draft auto-fill, legal-aid referrals, pre-filled eNothi/e-Gov forms where applicable, hearing & mediation tracker.

3) User Classes & Primary Personas

- **Citizen Reporter (web/mobile/USSD/WhatsApp):** submits reports, evidence, receives updates, rates outcomes.
- **Community Moderators (NGOs/CSOs):** verify/endorse incidents, escalate patterns, run micro-surveys.

- **Ward/Union Officer (LGD/City Corp):** receives cases, plans field visits, posts actions, closes cases.
- **Anti-Corruption Cell (ACC/LGD internal):** receives protected disclosures, runs data analytics, issues advisories.
- **Legal Aid Partner:** responds to legal queries, accepts referrals, manages case status.
- **Platform Admin (National/City):** configures jurisdictions, roles, workflows, SLAs, data sharing, and integrations.
- **Public Viewer:** sees anonymized maps, dashboards, budgets, and progress stories.

4) Top Use-Cases & User Journeys

1. **Fix-My-Street++:** Citizen logs a pothole with photos → AI auto-detects category and severity → assigns to Ward Engineer → SLA clock starts → contractor work order → photo proof of fix → citizen verifies → case closes; metrics contribute to public dashboard.
2. **Whistleblower (Pseudonymous):** Citizen alleges bribe solicitation → encrypted inbox + OTP alias → evidence locker with hash and timestamp → routed to ACC cell separate from LG workflows → status updates via code → optional reward schema after verified action.
3. **RTI/Legal Aid:** Wizard asks questions → generates a compliant RTI/complaint letter in Bangla/English → routes to proper authority → reminder schedule → legal aid referral if needed.
4. **Participatory Budgeting:** City opens a budget cycle → citizens propose, discuss, vote on neighborhood projects → moderators vet feasibility → council adopts shortlist → progress tracked on project cards.
5. **Disaster/Emergency Hotspot:** Surge in flood/drainage complaints → heatmap alerts → city task force coordination → resource dispatch → live map updates.

5) System Overview & Architecture (V1 microservices)

Frontend: SSR Laravel (existing) + REST/GraphQL API; Android (later PWA/Flutter).

Core Services:

- **Report/Case Service** (categories, evidence, geotag, dedupe, SLA engine, workflow states).
- **Workflow & SLA Engine** (per-jurisdiction configurable steps, timers, escalations, auto-reassign).
- **Identity & Access (IAM)** (roles/permissions; SSO adapters; pseudonymous mode).
- **Evidence Vault** (immutable hashes, chain-of-custody, redaction tools, access logs).
- **GIS/Geospatial Service** (tiling, clustering, jurisdiction boundaries, route planning).
- **Notification Service** (SMS, email, push, WhatsApp, in-app inbox).
- **Participation Suite** (polls/surveys, proposals, PB voting, town-hall Q&A).
- **Legal Aid & RTI Module** (guided flows, templates, referral directory, appointment slots).
- **Analytics & Open Data** (KPIs, heatmaps, trendlines, open datasets & APIs).
- **Audit & Transparency** (case logs, decision rationale, public status board).
- **Payments & Incentives (optional)** (micro-rewards/waivers; partner APIs like bKash/EkPay).

Integrations (adapters): eNothi/ACC intake (where permissible), City Corp ticketing, National ID verification (optional/consent-based), WhatsApp/Telegram bots, USSD/SMS gateway, cloud storage (S3/MinIO), map providers (GMAPS/OSM).

Security & Privacy: end-to-end transport encryption; evidence at-rest encryption; role-scoped access; redaction; whistleblower safe-mode; privacy impact assessment; data retention policies; incident response.

6) Functional Requirements (by module)

A) Report & Case Management

- Create report (geo, media, tags, anonymous/pseudonymous/identified).
- AI auto-classification (category, severity, jurisdiction, duplicate detection).
- Convert to case; assign owner; SLA start; status timeline; attachments; comments.
- Escalation rules; auto-reopen if citizen disagrees; 2-way messaging; field visit notes.
- Public transparency levels (private / anonymized public / fully public).

B) Anti-Corruption & Whistleblower

- Protected disclosure flow with OTP alias; encrypted mailbox; evidence vault.
- Risk triage: pattern detection, outlier vendors, repeat offenders, temporal spikes.
- Restricted dashboards for ACC/ethics units; outcome taxonomy (warning, suspension, etc.).

C) Legal & Justice Access

- Guided wizards: GD, complaint, RTI request, consumer protection, harassment reporting.
- Auto-generated letters/PDFs; checklist of required evidence; filing instructions.
- Partner directory (legal aid NGOs, govt help desks); referral tracking; appointment booking.

D) Local Governance & Participation

- Participatory budgeting: proposal intake, feasibility scoring, community voting, project tracking.
- Micro-surveys/polls (ward-level); town-hall agenda & minutes; public noticeboard.
- Ward performance dashboard: SLA compliance, backlogs, resolution rates, citizen satisfaction.

E) Community & Trust

- Endorse/verify reports by neighbors/CSOs; reputation for verifiers; misinformation controls.
- Story cards of successful fixes; before/after gallery; share to social media.

F) Mobile/Offline & Inclusion

- Bengali-first + English; low-bandwidth PWA; offline save-and-sync; USSD/SMS flows; accessibility (screen reader, high-contrast, font scaling).
- Kiosk mode for ward offices/community centers.

G) Admin & Ops

- Role-based access; jurisdiction setup; workflow builder; SLA templates; category taxonomy.
- Data policies; retention & deletion; redaction; export (CSV/JSON); open data portal controls.
- Notification templates; escalation ladders; holiday calendars.

7) Non-Functional Requirements

- **Performance:** <300ms p95 API latency; bulk queries paginated.
- **Scalability:** Multi-tenant (all city corps/municipalities); horizontal scaling; async workers.
- **Reliability:** 99.5% uptime; retry/queue for SMS/email; evidence checksum checks.
- **Security:** OWASP ASVS L2; audit logging; periodic pen-tests; KMS-managed keys.
- **Compliance:** Data-minimization; consent management; child-safety rules.

- **Observability:** structured logs; metrics; alerting; error tracking.

8) Data Model (high-level)

- **User**(id, role, jurisdiction_id, language, contact, KYC_status, reputation).
- **Report**(id, user_id?, geo, category, description, media[], created_at, privacy_level, source).
- **Case**(id, report_id, assignee_id, workflow_state, SLA_due, escalations[], actions[], resolution_summary).
- **Evidence**(id, case_id, file_uri, hash, redactions[], provenance, access_log).
- **Disclosure**(id, alias_id, channel, risk_score, status).
- **Proposal/BudgetItem, SurveyResponse, LegalReferral, NotificationLog, AuditLog.**

9) AI/ML Components (explainable)

- **Classifier:** category/severity/jurisdiction; confidence + rationale; human-in-the-loop.
- **Duplicate Detector:** perceptual hash (images), semantic text match, geo-proximity.
- **Anomaly/Corruption Analytics:** vendor anomaly, repeated bribe mentions, seasonality spikes.
- **Summarizer:** case briefs for officers; redaction suggestions; policy-aware templates.
- **Bias & Safety:** periodic drift tests; publish model cards; appeal pathway for users.

10) Integrations & APIs

- Open REST/GraphQL; webhook subscriptions; API keys per agency.

- Adapters: SMS/USSD gateway, WhatsApp bot, email, map tiles, payment providers.
- Optional: SSO with national identity (consent-based); eNothi ticket sync; ACC intake.

11) Governance, Roles & Moderation

- Clearly separated **public**, **civil society**, and **government** roles.
- Community Guidelines; moderation queues; CSO verification program.
- Transparency: case logs and decision rationales (where lawful).
- Ombudsperson workflow for appeals.

12) Gamification & Incentives

- Reputation for accurate reports and verifications.
- Badges for community champions; quarterly ward awards.
- Optional micro-rewards (vouchers/waivers) for verified fixes (policy-dependent).

13) Reporting, Dashboards & Open Data

- City/ward dashboards: inflow, backlog, SLA, median time-to-close, satisfaction.
- Heatmaps by category; seasonal trends; contractor performance league; corruption indicators.
- Open data portal: anonymized datasets; export tools; public API documentation.

14) Privacy, Safety & Ethics

- Pseudonymous mode by default for sensitive categories; doxxing safeguards.
- Evidence handling SOP; whistleblower protection practices; minimal exposure by design.
- Data classification (public/anonymized/confidential/restricted).
- Periodic ethics review with civil society board.

15) Mobile Apps & Channels

- PWA + Android (phase-2), with offline capture & background sync.
- WhatsApp/Telegram bot for quick reports and status checks.
- USSD tree for feature phones; SMS status codes.

16) Implementation Roadmap

MVP (4–6 weeks, for demo):

- Geo-tagged reports → case queue; SLA timers; status updates; public dashboard; SMS/email notifications; heatmap; basic AI classifier; pseudonymous submissions; RTI/complaint letter generator (1–2 flows); micro-survey tool; admin workflow builder (basic).

V1 (3–4 months):

- Evidence vault & chain-of-custody; duplicate detection; participatory budgeting beta; legal aid referrals; open data portal; WhatsApp bot; ward performance dashboards; escalation ladders.

V2 (6–9 months):

- Anti-corruption analytics suite; contractor performance analytics; mobile apps; USSD; multi-tenant rollout to 2–3 city corporations; advanced SSO; partner API marketplace.

17) KPIs & Impact Metrics

- Report→Resolution median days; % resolved within SLA; citizen satisfaction (CSAT) after closure; # active wards; # submissions from low-connectivity channels; # protected disclosures acted upon; % duplicate/false positives caught; reduction in repeat incidents in hotspots; budget project completion rate.

18) Risk Register & Mitigations

- **Retaliation risk:** pseudonymous default; redaction; restricted staff access; CSO escrow.
- **Data misuse:** role-based access; audit logs; DPA compliance; independent oversight.
- **Low adoption by agencies:** co-design workshops; SLA incentives; quick-wins on dashboards.
- **Model bias:** diverse training data; human-in-loop; appeals; transparency reports.
- **Connectivity barriers:** offline/PWA/USSD; kiosks; NGO partners.

19) Sustainability & Scale

- Multi-tenant SaaS for municipalities; open-source core with gov extensions; partner ecosystem (legal aid, NGOs, academic labs).
- Data governance MOUs; privacy policies; standard APIs for vendors.
- Revenue options (for sustainability): enterprise support contracts; analytics subscriptions for agencies; donor-funded deployments; CSR grants for community labs.

20) Application Pack — Ready-to-Fill Sections

- **Title:** Chokh-e-Dekha — Citizen-Powered Service Delivery & Anti-Corruption Platform.
- **Theme:** Civic Tech & Public Engagement (primary) + Local Governance & Participation + Anti-Corruption Tech & Data + Legal & Justice Access.

- **Problem & Context:** (Use §1)
- **Solution & Innovation:** (Use §§2–6, 9)
- **Beneficiaries:** Urban & rural citizens; city/municipal authorities; ACC; legal aid orgs.
- **Technology:** Microservices + AI + GIS + multi-channel (web, PWA, WhatsApp, USSD).
- **Prototype/Links:** (Screens + repo + demo DB).
- **Impact & KPIs:** (Use §17).
- **Roadmap:** (Use §16).
- **Team & Partners:** (Add city corp/NGO letters).
- **Data Protection & Ethics:** (Use §14).
- **Prize Utilization:** (Infra, pilots in 2 wards, USSD gateway, field training).

21) Screens & Artifacts to Prepare for Demo

- Citizen: report form (photo, map, privacy toggle), status page, feedback dialog.
- Officer: case queue with SLA, map clustering, action log, close case flow.
- Anti-corruption: protected inbox, evidence viewer, pattern dashboard.
- Legal aid: RTI/complaint wizard, referral inbox.
- Public: ward dashboard, heatmaps, participatory budget voting.

22) Appendix — Sample Data Schemas & Workflows

- JSON schemas for Report, Case, Evidence, Disclosure, Proposal.

- Workflow YAML for pothole fix and protected disclosure.
- Survey template (participation + satisfaction).
- RTI generator fields (authority, subject, facts, reliefs, attachments).

End of v1.0